

2. Suppose A is a 7×5 matrix with 5 pivots. Let $T(x) = Ax$ be a linear transformation from $\mathbb{R}^5$ into $\mathbb{R}^7$. Is T a one-to-one linear transformation? Is T onto $\mathbb{R}^7$?

Solution: Since there are only 5 pivots, it's obvious that there is ~~at least~~ ^{at least} one row having no pivots. By Theorem 4, there exists at least one $\hat{b}$ in $\mathbb{R}^7$ such that the equation $Ax = b$ has no solution. $\therefore$ For some b in $\mathbb{R}^7$, $T(x) = b$ has no solution. By defn, T is not an onto transformation.

By definition, a pivot position corresponds to the leading 1 of a row of a matrix in rref form. By the Row Reduction algorithm and Theorem 1, matrix A can be converted to a unique rref. From the defn. of rref, we know that each leading position of a non-zero row must be 1 and for each row, its leading entry must be in a column to the right of the columns of the leading entries of all rows above it.

$\therefore$ Each of the 5 columns are pivot columns. So in the echelon form of the augmented matrix $[A \ b]$ for some b in $\mathbb{R}^7$, there are no rows of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. $\therefore$ By Theorem 2, the system of equations is consistent. By defn, there are no free variables. $\therefore$ By Theorem 2, the corresponding system of equations has a unique solution.

$Ax = 0$ has only the unique trivial solution. $\therefore$ The columns of A are linearly independent. By Theorem 12(b), T is one-to-one. $\square$

Exercises 1.9

In Exercises 1-10, assume that T is a linear transformation. Find the standard matrix of T .

1. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$, $T(e_1) = (3, 1, 3, 1)$ and $T(e_2) = (-5, 2, 0, 0)$, where $e_1 = (1, 0)$ and $e_2 = (0, 1)$.

Solution: By Theorem 10, The standard matrix is $[T(e_1) \ T(e_2)]$

$$= \begin{bmatrix} 3 & -5 \\ 1 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}$$